EXTERNAL HALL / ABI: 3.3V or 5V logic -> 3.3V Schmitt buffer -> reset isolation

Joint feedback supply: fixed, protected 5V.
J601.1 = 5V output; J601.2/6 = GND. No power input.
For a 5V Hall/ABI encoder; no voltage selector.
Target sensor + pull-ups <=25mA. Fan has a separate branch on J502.

D601 rail isolated from supplies.
C604 is required; no 3V3 clamp tie.

INPUT OPTIONS
Default: 3.3V / 5V push-pull; pull-ups DNP.
Open-drain: fit R608-R610 (4.7k to protected 5V).
Do not fit pull-ups for externally driven push-pull.
100k pull-downs define disconnected inputs.
Single-ended only; common ground required.

D601
TPD3E001DRYR
NC
GND
VCC
IO3
IO2
IO1
FB_ESD_RAIL
C604
100nF 50V

J601
MP
MOUNT
EXT ENCODER / HALL

EXT_VCC
R608
4.7k DNP
R609
4.7k DNP
R610
4.7k DNP
R605
100R
R606
100R
R607
100R
R615
100k
R616
100k
R617
100k

U603A
U603C
U603B
74LVC3G17

U602A
TMUX1574PW
S1A
S1B
S2A
S2B
S3A
S3B
D1
D2
D3

TP1015
FEEDBACK_A_H1
TP1016
FEEDBACK_B_H2
FEEDBACK_J_H3

U604
TPS2553DBVR
IN
EN
ILIM
OUT
FAULT
GND
5V_SYS
R604
10k
EXT_POWER_FAULT_N
C606
4.7uF 16V X7R
C607
100nF 16V X7R

U604: ILIM tied to IN => 75mA nominal (50-100mA range).
IN and EN on 5V_SYS; continuous joint-feedback power.
EXT_POWER_FAULT_N -> PC13; wired-OR with fan limiter U503.
Stop motor if external feedback is lost; fault does not identify branch.

U602B
TMUX1574PW
VBUS_MUX_IN
VBUS_SENSE

U603D
74LVC3G17
C605
100nF

R305
10k
R306
10k
R307
10k
C603
100nF
INPUT_MUX_ENABLE_N
EXT_FEEDBACK_SELECT

U602C
TMUX1574PW
VDD
SEL
EN_N

Reset: mux disabled. PB8/BOOT0 held low by R307.
EN_N=0 enables VM sensing; wait >=2ms before ADC read.
SEL=0 grounds TIM4 inputs; SEL=1 connects external Hall/ABI.
Onboard absolute-angle SPI is independent of this switch.

3.3V / 5V external inputs; interface revision
ShiroFOC
Sheet: /External feedback/
File: 09_Feedback_Selection.kicad_sch
Title: External Hall / ABI conditioning and reset isolation
Size: A3 | Date: 2026-09-26 | **Rev: A1-Drawing**
KiCad E.D.A. 9.0.4 | Id: 10/12